

An all-orders two-replica proof of the complex weighted Krenn–Gu conjecture

September 26, 2026

Review status (2026-09-26): complete written argument; internally audited. This document gives the entire argument, including the earlier reflection and diagonal-reduction steps. Full Lean formalization and external peer review remain pending. It is a typeset presentation of the unchanged audited Markdown source, which retains its original review header. See the current review record for the internal audits and present status.

1 Statement and algebraic model

Theorem 1.1 (Complex weighted Krenn–Gu no-go). *Let V have even cardinality $n > 4$. At each vertex take three designated coordinates a, b, c . In the commutative algebra in which the product of any two coordinates at the same vertex is zero, let A be any quadratic involving only distinct vertices, with arbitrary complex coefficients. Write $A^{[j]} = A^j / j!$. Then*

$$A^{[n/2]} = \tau_a a^V + \tau_b b^V + \tau_c c^V, \quad \tau_a \tau_b \tau_c \neq 0 \tag{1}$$

is impossible. Here h^S means the product of coordinate h over S .

Coefficients of A are aggregate endpoint-color edge weights. Its divided top power is exactly the perfect-matching tensor: every matching occurs once. Parallel edges of identical endpoint colors aggregate by addition. Projecting any larger target palette onto three active colors preserves (1), so the result covers every target dimension at least three. Unequal nonzero pure amplitudes, arbitrary edge density, and arbitrary complex weights are allowed.

We use formal Wick moments, defined as sums over pairings of labeled factors with assigned symmetric covariances. Odd moments are zero. These are polynomial identities; they require neither positivity nor an invertible covariance. The proof of Theorem 1.1 is completed in Sections 5 and 6.

2 Whole binary higher responses vanish

Lemma 2.1 (Whole binary higher-response vanishing). *Let Ω have $2m + 1 \geq 3$ vertices, let R be an arbitrary quadratic on Ω , and let D be a vector space of rows L such that*

$$\Phi(L) = LR^{[m]} = \sum_h f_h(L) h^\Omega. \tag{2}$$

Assume all three linear functions f_h are nonzero and at least two are independent. For odd $p \leq 2m + 1$, define

$$H_p(L) = L^p R^{[(2m+1-p)/2]}.$$

For every $p \geq 3$, the tensor $H_p(L)$ has zero coefficient on every word using at most two colors, including the terminal degree $p = 2m + 1$. The same vanishing holds after polarization for products of arbitrary rows from the same D .

Proof. Give formal variables $X_{i,h}$ their R covariances, zero covariances within a vertex, and give an auxiliary g covariance $L_i(h)$ with $X_{i,h}$ and zero covariance with itself. A Wick moment with p copies of g and one specified coordinate at each vertex is the corresponding coefficient of $H_p(L)$. The $p!$ assignments of auxiliary occurrences are precisely those of the raw power L^p ; the divided power of R counts each remaining matching once.

Take two independent identical replicas (X, g_1) , (Y, g_2) . Choose distinct local colors α_i, β_i at each vertex and set

$$\Delta_i = X_{i,\alpha_i} Y_{i,\beta_i} - X_{i,\beta_i} Y_{i,\alpha_i}.$$

Over $\mathbb{C}(s, t)$, the orthogonal reflection

$$\frac{1}{s^2 + t^2} \begin{pmatrix} s^2 - t^2 & 2st \\ 2st & t^2 - s^2 \end{pmatrix}$$

fixes $sg_1 + tg_2$ and negates each Δ_i . It preserves the replicated covariance, so the moment of

$$(sg_1 + tg_2)^{p+1} \prod_{i \in \Omega} \Delta_i$$

is zero: there are an odd number of determinants. Extracting $s^p t$ gives

$$\sum_{I \subseteq \Omega} (-1)^{|I|} H_p(L)(w_I) \Phi(L)(w'_I) = 0, \quad (3)$$

where w_I uses β on I and α elsewhere, and w'_I is complementary. Equality over $\mathbb{C}(s, t)$ gives a polynomial identity, including isotropic parameter values by extension. No operation on the original source is being made.

With constant local pair a, b , (3) says

$$f_b[a^\Omega] H_p = f_a[b^\Omega] H_p.$$

Two independent f 's are coprime in $\mathbb{C}[D]$, hence for $p = 2r + 1$ there is a common homogeneous polynomial q_r of degree $2r$ such that $[h^\Omega] H_p = f_h q_r$ for every h ; $q_0 = 1$. For a mixed word w avoiding an active color h , choose $\alpha_i = w_i$, $\beta_i = h$. The only constant complementary word in (3) is h^Ω , giving $f_h[w] H_p = 0$. Polynomial cancellation proves all binary mixed coefficients zero, even at individual rows where $f_h(L) = 0$.

Define finite polynomials

$$\Psi(L) = \sum_{r=0}^m \frac{H_{2r+1}(L)}{(2r+1)!}, \quad g(L) = \sum_{r=0}^m \frac{q_r(L)}{(2r+1)!}.$$

On any binary palette a, b , $\Psi(L)$ thus equals

$$g(L)(f_a(L)a^\Omega + f_b(L)b^\Omega).$$

For two tensors on Ω , let K be the product of local alternating pairings, normalized by $K(a^\Omega, b^\Omega) = 1$. Wick moments with shifted variables $X_i + L_i$ show that $K(\Psi(L), \Psi(M))$ is invariant under simultaneous $\text{SO}(2)$ rotation of the two mean rows: the replica covariance and each determinant are unchanged. Choose independent f_a, f_b . This pairing is

$$(f_a(L)f_b(M) - f_b(L)f_a(M))g(L)g(M).$$

The determinant factor is a nonzero polynomial and is rotation invariant. Cancel it in $\mathbb{C}[D \times D]$, then rotate by 45 degrees and set $M = 0$. Since g is even and $g(0) = 1$,

$$g(L/\sqrt{2})^2 = g(L).$$

If g had positive degree, the degrees of the two sides would differ. Thus $g = 1$, and all q_r with $r \geq 1$ vanish. Together with (3), this proves whole binary vanishing through the terminal degree $2m + 1$. Polarization gives the same statement for products of arbitrary rows from this same D . $\square$

For a source (1), delete any root p . Its three actual incident rows have responses $\tau_h h^{V \setminus \{p\}}$, so their span satisfies (2) with three independent f_h . Thus Lemma 2.1 applies at every original root, before diagonalization.

3 Even omission identities and diagonal reduction

Lemma 3.1 (Even omission). *In the setting of (2), omit a vertex q , put $S = \Omega \setminus \{q\}$, and write*

$$\begin{aligned} R &= Q + \sum_h h_q V_h, & L &= U + \sum_h d_h h_q, \\ E(U) &= \sum_{r=0}^m \frac{U^{2r}}{(2r)!} Q^{[m-r]}, & F &= E(0). \end{aligned}$$

For any $L_1, L_2 \in D$ and any active color k ,

$$[k^S] U(L_1) U(L_2) Q^{[m-1]} = 0.$$

This includes $m = 1$.

Proof. The full coefficient of $q = h$ in $\Psi(L)$ is

$$d_h E(U) + E_{V_h}(U),$$

where subscripts mean directional derivatives in the mean, with Q fixed. By Lemma 2.1, its projection onto any palette h, k on S is $f_h(L)h^S$. All the following pairings use this binary projection implicitly.

Because $|S|$ is even, K is symmetric and $K(Z, h^S) = [k^S]Z$. The same two-replica rotation gives

$$\begin{aligned} K(E(U), E(U)) &= K(F, E(\sqrt{2}U)), \\ K(E(U), E_V(U)) &= \frac{1}{\sqrt{2}} K(F, E_V(\sqrt{2}U)). \end{aligned} \tag{4}$$

The second identity follows by differentiating the rotation identity at means $U, U + tV$: the second rotated mean is zero at $t = 0$ and $E'(0) = 0$. Substitute

$$E_{V_h}(U) = f_h h^S - d_h E(U)$$

and the corresponding identity for $\sqrt{2}L$ into the second equation of (4). The first cancels the d_h terms, leaving

$$f_h(L)[k^S](E(U) - F) = 0.$$

Cancel the nonzero polynomial f_h . Separating homogeneous degrees and polarizing yields

$$[k^S] U(L_1)U(L_2)Q^{[m-1]} = 0. \quad (5)$$

□

Lemma 3.2 (Global diagonal reduction). *Every endpoint-color edge block of a source (1) is diagonal in the given target coordinates.*

Proof. For a fixed color h , define

$$\begin{aligned} M_h[p, r] &= A_{pr}(h, h) & (p \neq r), & \quad M_h[p, p] = 0, \\ C_h[r, q] &= \text{haf}(M_h[V \setminus \{r, q\}]) & (r \neq q), & \quad C_h[q, q] = 0, \\ B_{ih}[p, r] &= A_{pr}(i, h) & (p \neq r), & \quad B_{ih}[p, p] = 0. \end{aligned}$$

Expansion of the original one-defect word at p gives

$$(B_{ih}C_h)[p, p] = \delta_{i,h}\tau_h.$$

For $p \neq q$, expand each $C_h[r, q]$ at its retained vertex p :

$$\begin{aligned} (B_{ih}C_h)[p, q] &= \sum_{\substack{r, s \in V \setminus \{p, q\} \\ r \neq s}} A_{pr}(i, h) A_{ps}(h, h) \text{haf}(M_h[V \setminus \{p, q, r, s\}]) \\ &= 0. \end{aligned} \quad (6)$$

The last expression is exactly (5) for two rows of the original p family after omitting q . Both sums are ordered; there is no factor $1/2$. Consequently

$$B_{ih}C_h = \delta_{i,h}\tau_h I.$$

Taking $i = h$ proves C_h invertible. Taking $i \neq h$ then proves $B_{ih} = 0$. Every original edge block is therefore diagonal. This uses neither cofactor nonvanishing assumptions nor a change of basis. □

4 The endpoint identity

Fix two colors B, H after diagonal reduction and roots $p \neq q$. Put $U = V \setminus \{p, q\}$, $N = |U|$ (even), and let Q be the literal binary quadratic on U . Write

$$\begin{aligned} d &= B_{pq}, & e &= H_{pq}, & \beta &= \text{haf}(B), \\ \eta &= \text{haf}(H), & \alpha &= \text{haf}(B[U]), \\ x &= \sum_{i \in U} B_{pi}b_i, & y &= \sum_{i \in U} B_{qi}b_i, \\ u &= \sum_{i \in U} H_{pi}h_i, & v &= \sum_{i \in U} H_{qi}h_i, \\ E(L) &= \sum_{j=0}^{N/2} \frac{L^{2j}}{(2j)!} Q^{[N/2-j]}, & F &= E(0). \end{aligned}$$

Theorem 4.1 (Supported endpoint identity). *For every supported edge $B_{pq} \neq 0$,*

$$\beta = B_{pq} \operatorname{haf}(B[V \setminus \{p, q\}]).$$

Derivatives of E always keep Q fixed. Lemma 2.1, applied to the original p and q row planes and expanded at the other root, gives the exact polynomial identities

$$\begin{aligned} E_y(sx + tu) + dsE(sx + tu) &= s\beta b^U, \\ E_v(sx + tu) + etE(sx + tu) &= t\eta h^U, \\ E_x(ay + bv) + daE(ay + bv) &= a\beta b^U, \\ E_u(ay + bv) + ebE(ay + bv) &= b\eta h^U. \end{aligned} \tag{7}$$

These include the highest coefficient, which uses the terminal original odd response of degree $N + 1$. They are identities in mean parameters on the unchanged source. Differentiating those parameters is legitimate and gives

$$\begin{aligned} E_u(ay) &= 0, & E_v(sx) &= 0, \\ E_{yu}(sx) &= -dsE_u(sx), \\ E_{uv}(sx) + eE(sx) &= \eta h^U, \\ E_{xy}(0) + dF &= \beta b^U. \end{aligned} \tag{8}$$

On tensors on U , use the symmetric binary alternating pairing K , normalized by $K(b^U, h^U) = 1$. The polynomial $K(E(L_1), E(L_2))$ is invariant under the full $O(2)$ action on the two means. The Wick proof is as in Section 2; a determinant -1 transformation contributes $(-1)^N = 1$.

Introduce independent columns $P, Q_c, R_c, S_c \in \mathbb{C}^2$ and means $L_i = P_i x + (Q_c)_i y + (R_c)_i u + (S_c)_i v$. Set

$$\begin{aligned} G &= K(E(L_1), E(L_2)), & f &= G|_{R_c=S_c=0}, \\ M_{ij} &= \partial_{(R_c)_i} \partial_{(S_c)_j} G|_{R_c=S_c=0}, & T &= M + efI. \end{aligned}$$

These are finite polynomials. Orthogonal invariance gives

$$M(OP, OQ_c) = OM(P, Q_c)O^T,$$

and similarly for T . By (8),

$$M_{12} = K(E_u(P_1 x + (Q_c)_1 y), E_v(P_2 x + (Q_c)_2 y))$$

is divisible by $P_1(Q_c)_2$; likewise M_{21} is divisible by $P_2(Q_c)_1$.

Lemma 4.2 (Polynomial orthogonal kernel normal form). *For the kernel matrix just defined,*

$$\begin{aligned} T &= aI + bPQ_c^T, \\ a, b &\in \mathbb{C}[\sigma, \tau, c], \\ \sigma &= P \cdot P, & \tau &= Q_c \cdot Q_c, & c &= P \cdot Q_c. \end{aligned} \tag{9}$$

Proof. In an orthonormal frame where $P_1 = 0$, $M_{12} = 0$, so MP is parallel to P . In a frame where $(Q_c)_2 = 0$, $M_{12} = 0$, so $M^T Q_c$ is parallel to Q_c . Such frames exist on a dense open set of

nonisotropic columns. Their eigenvalues agree when $P \cdot Q_c \neq 0$. For two independent columns in dimension two, these conditions force

$$M = a_0 I + b P Q_c^T.$$

Indeed, M minus the common eigenvalue times I annihilates P and has left kernel Q_c ; its rank-one form is a multiple of $(P \cdot Q_c)I - P Q_c^T$. Literal divisibility proves that

$$b = \frac{M_{12}}{P_1(Q_c)_2}, \quad a_0 = M_{11} - b P_1(Q_c)_1$$

are polynomials. The generic identity therefore extends everywhere. Uniqueness and covariance make a_0, b scalar $O(2)$ invariants, and the same is true of $a = a_0 + e f$.

For completeness, the invariant ring assertion is elementary. In coordinates $p_+ = P_1 + i P_2$, $p_- = P_1 - i P_2$ and q_+, q_- , $SO(2)$ acts with weights $+1, -1$. Its invariant monomials are generated by $\sigma, \tau, r = p_+ q_-, s = p_- q_+$, with $rs = \sigma \tau$. Reflection exchanges r, s . Symmetric polynomials reduce to $r + s = 2c$ and $rs = \sigma \tau$, giving $\mathbb{C}[\sigma, \tau, c]$. These three Gram coordinates are algebraically independent. $\square$

Proof of Theorem 4.1. At $(Q_c)_1 = 0$, differentiate M_{12} in $(Q_c)_1$ and use (8). One obtains

$$\partial_{(Q_c)_1} M_{12} \Big|_{(Q_c)_1=0} = -d P_1 M_{12} \Big|_{(Q_c)_1=0}, \quad P_1^2(Q_c)_2(b_c + db) = 0.$$

The substitution

$$(\sigma, \tau, c) = (P_1^2 + P_2^2, (Q_c)_2^2, P_2(Q_c)_2)$$

is injective on polynomial rings: every $\tau \neq 0$ and arbitrary σ, c is attained over $\mathbb{C}$. Thus $b_c + db = 0$. Since $d \neq 0$, comparison of the highest power of c forces the finite polynomial b to be zero. Therefore $T = aI$.

Write

$$Z = E_{uv}(P_2 x + (Q_c)_2 y) + e E(P_2 x + (Q_c)_2 y).$$

The full expression is $T_{22} = K(E(P_1 x + (Q_c)_1 y), Z)$. Differentiate this expression in $(Q_c)_1$, then set $(Q_c)_1 = 0$. Equation (7) gives

$$\partial_{(Q_c)_1} T_{22} \Big|_{(Q_c)_1=0} = P_1 \beta K(b^U, Z) - d P_1 T_{22} \Big|_{(Q_c)_1=0}.$$

The pure- H coefficient of Z is η : positive B mean insertions cannot contribute to an all- H word, so it is the same as at zero, where (8) gives $E_{uv}(0) + e F = \eta h^U$. Thus $K(b^U, Z) = \eta$. Using (9) and the same injective substitution proves

$$a_c + da = \beta \eta.$$

Finite polynomial degree and $d \neq 0$ force $a = \beta \eta / d$, independent of all three Gram variables. At the origin,

$$a(0) = K(F, E_{uv}(0) + e F) = \eta K(F, h^U) = \eta \alpha.$$

Since $\eta \neq 0$, this proves

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]) \quad \text{whenever } B_{pq} \neq 0. \quad (10)$$

No source deformation, differentiation of source equations along an unlicensed direction, infinite series, or positivity argument occurs here. The variables P, Q_c are independent proof parameters, not actual graph rows. $\square$

5 Every color is a perfect matching

Corollary 5.1 (Matching rigidity of every active color). *Each color graph of a source (1) is a weighted perfect matching.*

Proof. The ordinary hafnian expansion at a vertex p says

$$\beta = \sum_{q \neq p} B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

By (10), every nonzero summand equals β . Since $\beta \neq 0$, the number of supported B neighbors of p is exactly one. This holds at every vertex, and for each of the three colors by choosing any other active color as H . $\square$

Two such matchings cannot share an edge pq : choosing that edge in color B and the other matching in color H off p, q gives a mixed word with nonzero coefficient, the product of nonzero matching weights. Consequently their union is a simple cubic graph properly colored with three colors. Any receiving color word supports at most one matching, because each vertex has exactly one incident edge of its requested color. Therefore a mixed perfect matching would have a nonzero coefficient and violate (1).

6 The remaining graph lemma

The following standard unweighted obstruction is usually attributed to Bogdanov; it is also stated as Theorem 7 in the HTML version of Chandran–Gajjala–Illickan, Krenn–Gu conjecture for sparse graphs. No new graph-theoretic result is claimed here.

Lemma 6.1 (Three-matching obstruction). *Three pairwise disjoint perfect matchings F, G, H on $n > 4$ vertices have a mixed perfect matching in their union.*

Proof. Suppose otherwise. The union $F \cup G$ must be a single alternating Hamilton cycle C : otherwise choosing F on one component and G on the others already gives a mixed matching. Label its vertices $0, \dots, n-1$ cyclically. The matching H consists of chords.

A chord with endpoints of opposite parity leaves two even paths when its endpoints are deleted. Matching those paths along C and adding the chord gives a mixed perfect matching. Hence all H chords join equal parities. The two parity classes must each have even size; otherwise H is impossible.

An even-even chord and an odd-odd chord whose endpoints interlace leave four even paths on deleting their endpoints. Those paths and the two chords give a mixed perfect matching when $n > 4$. Thus no such pair interlaces.

Choose a chord and one of its sides with the fewest vertices strictly inside, over all chords and both sides. Its endpoints have equal parity, so its interior contains a vertex of the opposite parity. Every vertex of that opposite parity in the interior must be H -matched to another interior vertex; a partner outside would give an interlacing opposite-parity chord. Therefore some H chord lies strictly inside the chosen arc. One side of that chord has fewer interior vertices, a contradiction. $\square$

This supplies a mixed perfect matching, contradicting Section 5. Equation (1) is impossible for every even $n > 4$, completing the proof of Theorem 1.1.

7 Boundaries, provenance, and review scope

For $n = 4$, the three matchings of K_4 give the permitted ternary source. In Section 6, a pair of interlacing chords then exhausts the vertices and gives the pure third matching, so no contradiction is asserted. Sections 2–5 remain valid at $n = 4$, including their terminal response coefficients. For $n = 2$, the odd-core lemma does not apply and arbitrary target dimension is possible. Odd n has no perfect matchings in this model.

The earlier written foundations are collected in the higher-response directory, with separate proofs and audits of reflection, even omission, and global diagonal reduction. The new step is Section 4: finite $O(2)$ covariance and two original root identities force the endpoint equality. Sections 2–3 are included here so that the all-orders conclusion has no dependency on a finite enumeration, an unproved normalization operation, or an assumed vanishing of three-color higher responses.